1. Investigating pertussis cases by year
2. A tale of two vaccines (wP & aP)
3. Exploring CMI-PB data
4. Examine IgG Ab titer levels
5. CMI-PB RNA-Seq data
Session info

Lab 18: Investigating Pertussis Resurgence Mini-Project

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```r
library(ggplot2)
library(dplyr)
library(jsonlite)
library(lubridate)
library(tidyr)
```

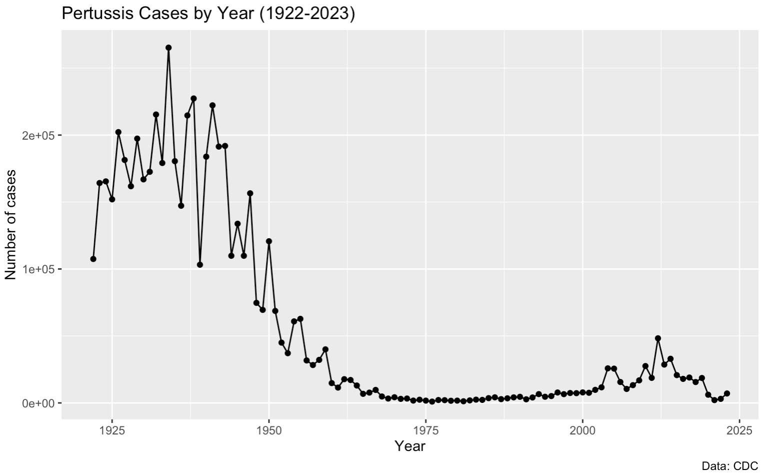
1. Investigating pertussis cases by year

Pertussis (whooping cough) is a highly contagious respiratory disease caused by *Bordetella pertussis*. The CDC has been tracking US case counts since 1922 through the National Notifiable Diseases Surveillance System (NNDSS).

Q1. Make a `cdc` data frame and plot cases over time

```r
cdc <- data.frame(
  year = 1922:2023,
  cases = c(107473,164191,165418,152003,202210,181411,161799,197371,166914,
            172559,215343,179135,265269,180518,147237,214652,227319,103188,
            183866,222202,191383,191890,109873,133792,109860,156517,74715,
            69479,120718,68687,45030,37129,60886,62786,31732,28295,32148,
            40005,14809,11468,17749,17135,13005,6799,7717,9718,4810,3285,
            4249,3036,3287,1759,2402,1738,1010,2177,2063,1623,1730,1248,
            1895,2463,2276,3589,4195,2823,3450,4157,4570,2719,4083,6586,
            4617,5137,7796,6564,7405,7298,7867,7580,9771,11647,25827,25616,
            15632,10454,13278,16858,27550,18719,48277,28639,32971,20762,
            17972,18975,15609,18617,6124,2116,3044,7063)
)

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases by Year (1922-2023)",
       x = "Year", y = "Number of cases",
       caption = "Data: CDC")
```


Pertussis Cases by Year (1922-2023)

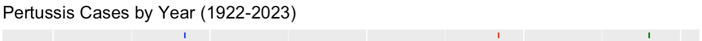
Data: CDC

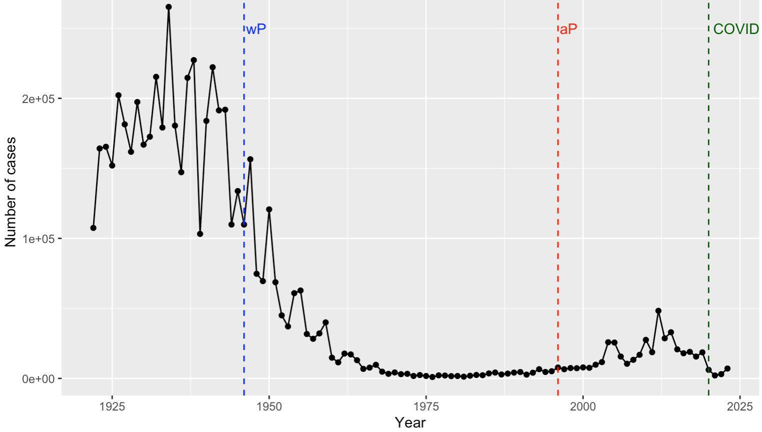
2. A tale of two vaccines (wP & aP)

Two pertussis vaccines have been used in the US: the older whole-cell (wP) vaccine introduced in 1946, and the acellular (aP) vaccine which replaced it in 1996. The aP vaccine uses only purified antigens of the bacterium and was developed to reduce the side effects of wP.

Q2. Add vlines for the 1946 wP introduction and the 1996 aP switch. What do you notice?

```r
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, color = "blue",  linetype = "dashed") +
  geom_vline(xintercept = 1996, color = "red",   linetype = "dashed") +
  geom_vline(xintercept = 2020, color = "darkgreen", linetype = "dashed") +
  annotate("text", x = 1946, y = 250000, label = "wP",  color = "blue",  hjust = -0.1) +
  annotate("text", x = 1996, y = 250000, label = "aP",  color = "red",   hjust = -0.1) +
  annotate("text", x = 2020, y = 250000, label = "COVID", color = "darkgreen", hjust = -0.1) +
  labs(title = "Pertussis Cases by Year (1922-2023)",
       x = "Year", y = "Number of cases")
```


Pertussis Cases by Year (1922-2023)



After the wP vaccine was introduced in 1946, cases dropped dramatically — from over 200,000 per year in the 1940s to under 2,000 per year by the late 1970s — and stayed low through the early 1990s. After the 1996 switch to aP, cases began climbing again. The resurgence peaked at 48,277 cases in 2012, the highest annual total since 1955. The 2020–2021 dip lines up with COVID-era reduced transmission and reporting, and cases are starting to climb again by 2023.

Q3. Describe what happened after the introduction of the aP vaccine. Possible explanation?

Despite high aP coverage, US pertussis cases are once again increasing, with the largest outbreak since the pre-vaccine era happening in 2012. There are several non-mutually-exclusive hypotheses for the resurgence:

1. **Waning immunity from aP** — aP-induced protection appears to be shorter-lived than wP-induced protection, so adolescents originally primed as infants with aP lose immunity sooner. This is the leading hypothesis and is exactly what the CMI-PB project is set up to investigate.
2. **More sensitive diagnostics** — PCR-based testing detects cases that older culture-based methods would have missed.
3. **Vaccination hesitancy** — lower coverage reduces herd immunity.
4. **Bacterial evolution** — *B. pertussis* strains may be evolving to escape vaccine-induced immunity (for example, the rise of pertactin-deficient strains).

3. Exploring CMI-PB data

The CMI-PB project tracks long-term immune responses in individuals who received either DTwP or DTaP vaccines in infancy and then a Tdap booster as adults. A booster acts as a proxy for what happens during a natural infection. CMI-PB provides this data via a JSON API.

```
subject  <- read_json("https://www.cmi-pb.org/api/subject",  simplifyVector = TRUE)
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer    <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)

head(subject, 3)
```

```
##   subject_id infancy_vac biological_sex              ethnicity  race
## 1          1          wP         Female Not Hispanic or Latino White
## 2          2          wP         Female Not Hispanic or Latino White
## 3          3          wP         Female                Unknown White
##   year_of_birth date_of_boost      dataset
## 1    1986-01-01    2016-09-12 2020_dataset
## 2    1968-01-01    2019-01-28 2020_dataset
## 3    1983-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy-vaccinated subjects?

```
table(subject$infancy_vac)
```

```
## 
## aP wP 
## 87 85
```

Q5. How many Male and Female subjects?

```
table(subject$biological_sex)
```

```
## 
## Female   Male 
##    112     60
```

Q6. Breakdown of race and biological sex

```
table(subject$race, subject$biological_sex)
```

```
##                                            
##                                             Female Male
##   American Indian/Alaska Native                  0    1
##   Asian                                         32   12
##   Black or African American                      2    3
##   More Than One Race                            15    4
##   Native Hawaiian or Other Pacific Islander      1    1
##   Unknown or Not Reported                       14    7
##   White                                         48   32
```

Side-note: working with dates with lubridate

```
today()
```

```
## [1] "2026-06-01"
```

```
today() - ymd("2000-01-01")
```

```
## Time difference of 9648 days
```

```
time_length(today() - ymd("2000-01-01"), "years")
```

```
## [1] 26.41478
```

Q7. Average age of wP vs aP individuals, and are they significantly different?

```
subject$age <- today() - ymd(subject$year_of_birth)

ap <- subject %>% filter(infancy_vac == "aP")
wp <- subject %>% filter(infancy_vac == "wP")

round(summary(time_length(ap$age, "years")))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23      27      28      28      29      35
```

```
round(summary(time_length(wp$age, "years")))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23      33      35      37      40      58
```

```
t.test(time_length(wp$age, "years"),
       time_length(ap$age, "years"))$p.value
```

```
## [1] 2.372101e-23
```

aP subjects are in their mid-20s and wP subjects are in their mid-30s. The p-value is vanishingly small, so the two groups are significantly different in age — which makes sense given the timing of the vaccine switch.

Q8. Age of all individuals at time of boost

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

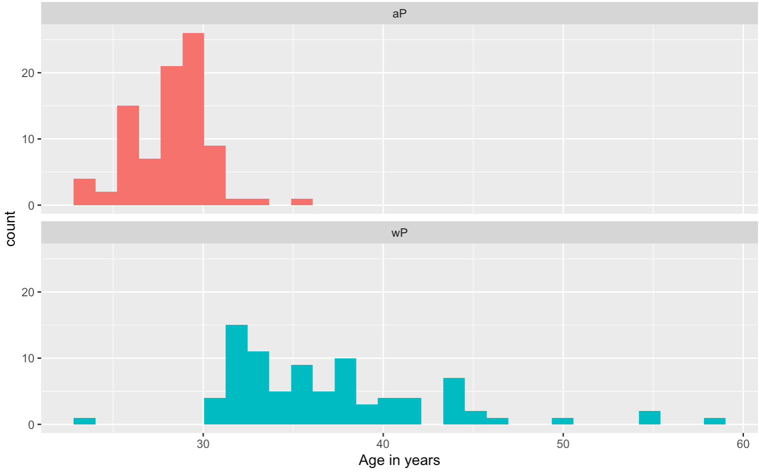
```
## [1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

```
summary(age_at_boost)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.83   21.03   25.75   26.09   29.56   51.07
```

Q9. Are the two groups significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"), fill = as.factor(infancy_vac)) +
  geom_histogram(show.legend = FALSE, bins = 30) +
  facet_wrap(vars(infancy_vac), nrow = 2) +
  xlab("Age in years")
```



The two histograms barely overlap, and combined with the very small p-value above, yes — the wP and aP groups are clearly different in age.

Joining tables

The CMI-PB tables share common ID columns. To know whether a `specimen_id` came from an aP or wP subject, we need to join the `specimen` table to the `subject` table.

Q9 (joining). Join `specimen` and `subject` into `meta`

```
meta <- inner_join(specimen, subject)
dim(meta)
```

```
## [1] 1503   14
```

```
head(meta)
```

```
##   specimen_id subject_id actual_day_relative_to_boost
## 1           1          1                           -3
## 2           2          1                            1
## 3           3          1                            3
## 4           4          1                            7
## 5           5          1                           11
## 6           6          1                           32
##   planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1                             0         Blood     1          wP         Female
## 2                             1         Blood     2          wP         Female
## 3                             3         Blood     3          wP         Female
## 4                             7         Blood     4          wP         Female
## 5                            14         Blood     5          wP         Female
## 6                            30         Blood     6          wP         Female
##                ethnicity  race year_of_birth date_of_boost      dataset
## 1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
##          age
## 1 14761 days
## 2 14761 days
## 3 14761 days
## 4 14761 days
## 5 14761 days
## 6 14761 days
```

Q10. Join `meta` and `titer` into `abdata`

```
abdata <- inner_join(titer, meta)
dim(abdata)
```

```
## [1] 52576    21
```

Q11. How many specimens per isotype?

```
table(abdata$isotype)
```

```
## 
##   IgE   IgG  IgG1  IgG2  IgG3  IgG4 
##  6698  5389 10117 10124 10124 10124
```

Q12. Different `$dataset` values, and what's notable about the most recent one?

```
table(abdata$dataset)
```

```
## 
## 2020_dataset 2021_dataset 2022_dataset 2023_dataset 
##        31520         8085         7301         5670
```

The most recent dataset has the fewest rows. CMI-PB is an ongoing longitudinal study, so the newest collection year has fewer subjects fully processed and fewer visits completed than the older 2020/2021 datasets — data accumulates over time.

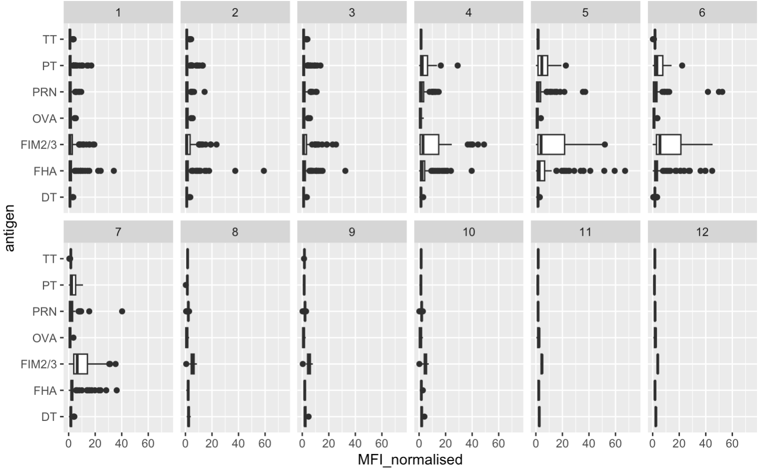
4. Examine IgG Ab titer levels

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

```
##   specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
## 1           1     IgG                TRUE      PT   68.56614       3.736992
## 2           1     IgG                TRUE     PRN  332.12718       2.602350
## 3           1     IgG                TRUE     FHA 1887.12263      34.050956
## 4          19     IgG                TRUE      PT   20.11607       1.096366
## 5          19     IgG                TRUE     PRN  976.67419       7.652635
## 6          19     IgG                TRUE     FHA   60.76626       1.096457
##    unit lower_limit_of_detection subject_id actual_day_relative_to_boost
## 1 IU/ML                 0.530000          1                           -3
## 2 IU/ML                 6.205949          1                           -3
## 3 IU/ML                 4.679535          1                           -3
## 4 IU/ML                 0.530000          3                           -3
## 5 IU/ML                 6.205949          3                           -3
## 6 IU/ML                 4.679535          3                           -3
##   planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1                             0         Blood     1          wP         Female
## 2                             0         Blood     1          wP         Female
## 3                             0         Blood     1          wP         Female
## 4                             0         Blood     1          wP         Female
## 5                             0         Blood     1          wP         Female
## 6                             0         Blood     1          wP         Female
##                ethnicity  race year_of_birth date_of_boost      dataset
## 1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
## 4                Unknown White    1983-01-01    2016-10-10 2020_dataset
## 5                Unknown White    1983-01-01    2016-10-10 2020_dataset
## 6                Unknown White    1983-01-01    2016-10-10 2020_dataset
##          age
## 1 14761 days
## 2 14761 days
## 3 14761 days
## 4 15857 days
## 5 15857 days
## 6 15857 days
```

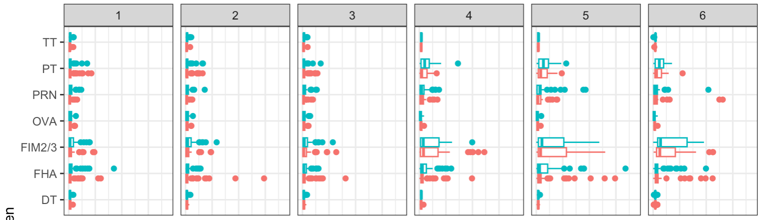
Q13. Boxplot of MFI for all antigens, faceted by visit

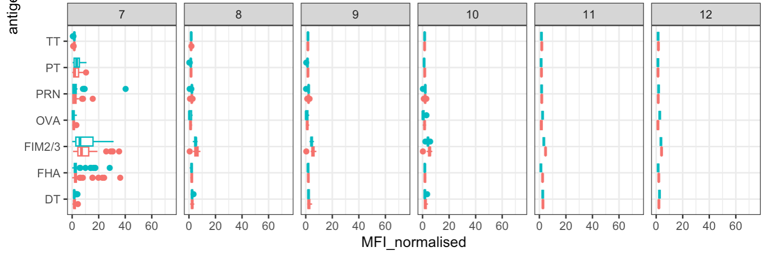
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2)
```



Q14. Which antigens differ in IgG titers over time, and why?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow = 2) +
  xlim(0, 75) +
  theme_bw()
```



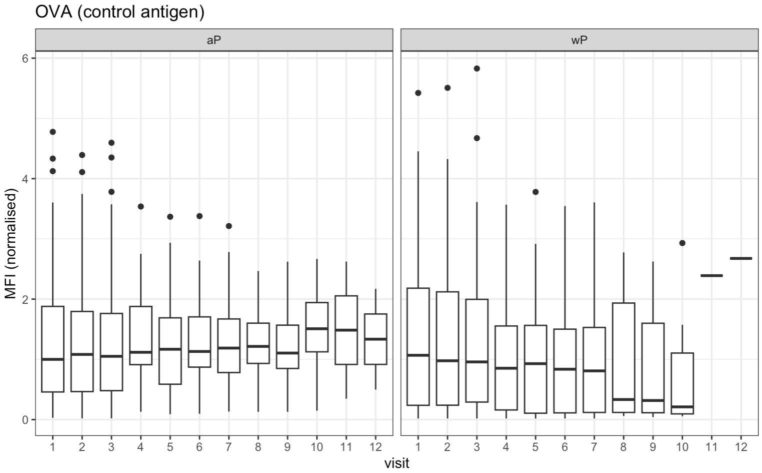


PT, FHA, FIM2/3, and PRN show the clearest rises in IgG MFI across visits. These are exactly the antigens included in the aP/Tdap vaccines, so the booster drives a measurable antibody response against them. The control antigen OVA (chicken ovalbumin — not in any pertussis vaccine) stays flat, and other non-vaccine antigens like BETV1 and LOLP1 do too.

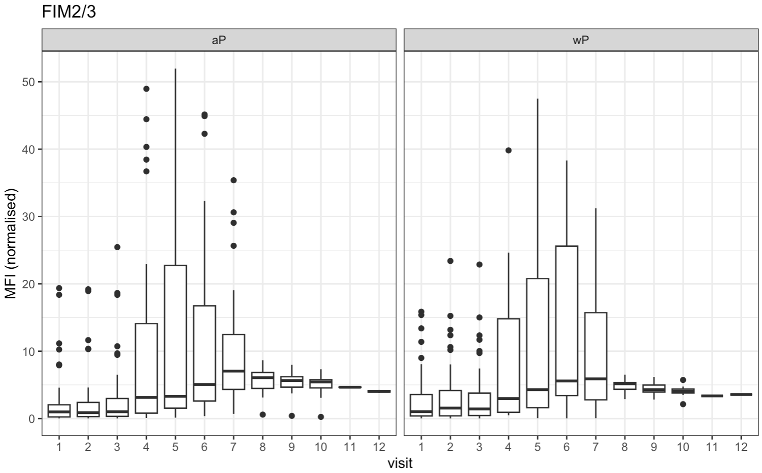
Q15. Boxplots for selected antigens

Looking at PT (a key pertussis virulence factor), FIM2/3 (another vaccine antigen), and OVA (control) across visits, faceted by `infancy_vac`:

```
filter(igg, antigen == "OVA") %>%
  ggplot() +
  aes(as.factor(visit), MFI_normalised) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(infancy_vac)) +
  labs(title = "OVA (control antigen)", x = "visit", y = "MFI (normalised)") +
  theme_bw()
```

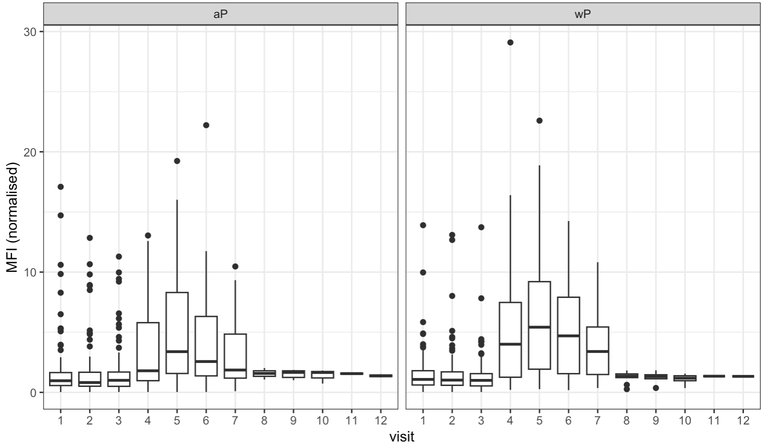


```
filter(igg, antigen == "FIM2/3") %>%
  ggplot() +
  aes(as.factor(visit), MFI_normalised) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(infancy_vac)) +
  labs(title = "FIM2/3", x = "visit", y = "MFI (normalised)") +
  theme_bw()
```



```
filter(igg, antigen == "PT") %>%
  ggplot() +
  aes(as.factor(visit), MFI_normalised) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(infancy_vac)) +
  labs(title = "PT (pertussis toxin)", x = "visit", y = "MFI (normalised)") +
  theme_bw()
```

PT (pertussis toxin)



Q16. What do you notice about these time courses, and PT in particular?

OVA stays flat across all visits in both groups, which is what we'd expect for an antigen that isn't in the vaccine. PT and FIM2/3 both rise sharply after the Tdap boost, peak around visit 5, and then decline. PT in particular shows the largest fold-change. The shape of the response looks similar between wP and aP groups.

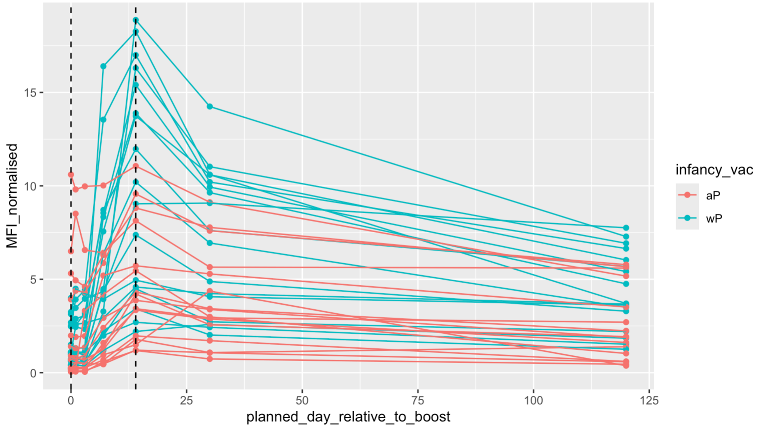
Q17. Any clear aP vs wP differences?

Both groups mount a strong booster response with similar peak timing (visit 5) and a similar overall shape. wP subjects sometimes reach modestly higher peaks for PT and FHA, but the difference is not dramatic at this resolution. The hypothesized wP vs aP difference is in the *durability* of the response and possibly in qualitative immune differences (cellular response, IgG subclass distribution) rather than in the peak MFI.

2021 dataset IgG PT time course

```r
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
  aes(x = planned_day_relative_to_boost,
      y = MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 0,  linetype = "dashed") +
  geom_vline(xintercept = 14, linetype = "dashed") +
  labs(title    = "2021 dataset IgG PT",
       subtitle = "Dashed lines: day 0 (pre-boost) and day 14 (apparent peak)")
```


2021 dataset IgG PT

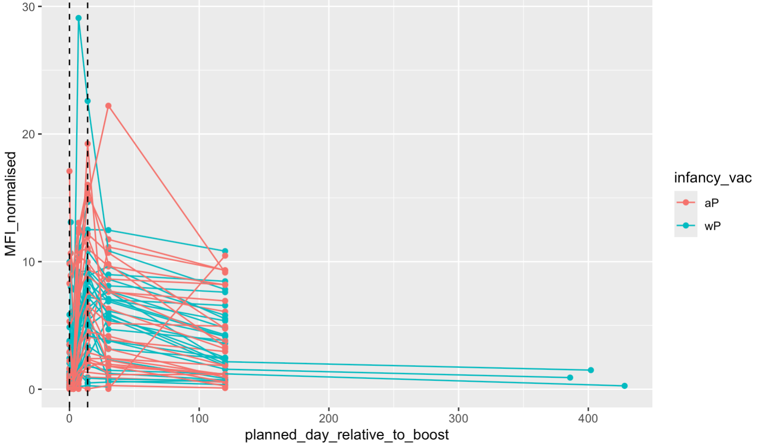
Dashed lines: day 0 (pre-boost) and day 14 (apparent peak)

Q18. Does this trend look similar for the 2020 dataset?

```r
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
  aes(x = planned_day_relative_to_boost,
      y = MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 0,  linetype = "dashed") +
  geom_vline(xintercept = 14, linetype = "dashed") +
  labs(title = "2020 dataset IgG PT")
```

2020 dataset IgG PT



Yes, same pattern — IgG PT is low pre-boost, spikes by day 14, and then declines. The 2020 dataset has a longer follow-up window, so the decay phase after the peak is more visible.

5. CMI-PB RNA-Seq data

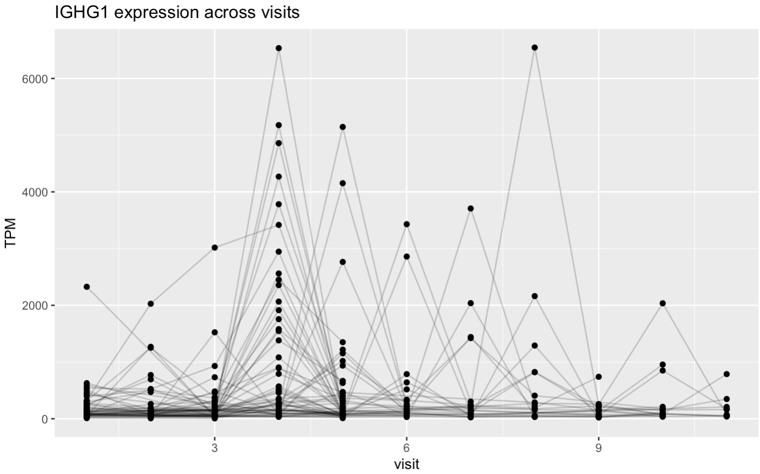
The CMI-PB API also exposes RNA-Seq data. We can pull expression for a specific gene via a versioned ENSEMBL ID. Here we look at IGHG1, the gene encoding the constant region of the IgG1 antibody heavy chain.

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)

ssrna <- inner_join(rna, meta)
```

Q19. Time course of IGHG1 expression

```
ggplot(ssrna) +
  aes(visit, tpm, group = subject_id) +
  geom_point() +
  geom_line(alpha = 0.2) +
  labs(title = "IGHG1 expression across visits", y = "TPM")
```



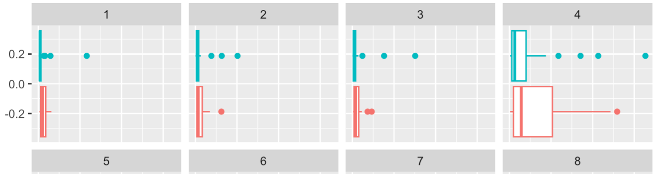
Q20. When is IGHG1 expression at its maximum?

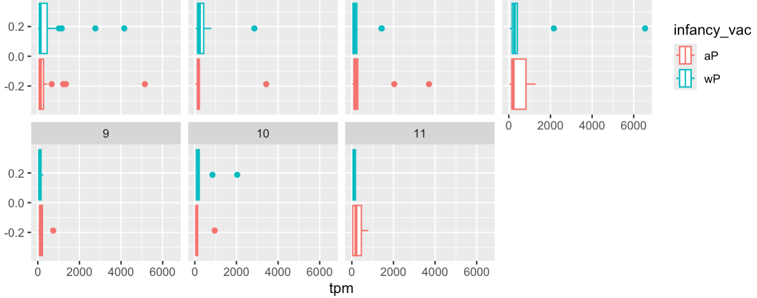
Expression peaks around visit 4 (a few days post-boost) and then drops back toward baseline by visit 5–6.

Q21. Does this pattern match the antibody titer trend? If not, why not?

No — the RNA-Seq peak at visit 4 *precedes* the antibody peak at visit 5. This makes sense biologically: the cells ramp up IGHG1 transcription first, but the IgG protein then needs to be translated, assembled, secreted, and accumulate in the blood, which takes additional time. Antibodies are also long-lived, so MFI stays elevated long after the transcript has fallen back to baseline.

```
ggplot(ssrna) +
  aes(tpm, col = infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```





There’s no obvious wP vs aP difference at the IGHG1 transcript level.

Session info

```
sessionInfo()
```

```
## R version 4.5.3 (2026-03-11)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5
## 
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib 
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
ersion 3.12.1
## 
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
## 
## time zone: America/Los_Angeles
## tzcode source: internal
## 
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base     
## 
## other attached packages:
## [1] tidyr_1.3.2     lubridate_1.9.5 jsonlite_2.0.0  dplyr_1.2.1    
## [5] ggplot2_4.0.3  
## 
## loaded via a namespace (and not attached):
##  [1] vctrs_0.7.3        cli_3.6.6          knitr_1.51         rlang_1.2.0       
##  [5] xfun_0.57          otel_0.2.0         purrr_1.2.2        generics_0.1.4    
##  [9] S7_0.2.2           labeling_0.4.3     glue_1.8.1         htmltools_0.5.9   
## [13] sass_0.4.10        scales_1.4.0       rmarkdown_2.31     grid_4.5.3        
## [17] tibble_3.3.1       evaluate_1.0.5     jquerylib_0.1.4    fastmap_1.2.0     
## [21] yaml_2.3.12        lifecycle_1.0.5    compiler_4.5.3     RColorBrewer_1.1-3
## [25] timechange_0.4.0   pkgconfig_2.0.3    rstudioapi_0.18.0  farver_2.1.2      
## [29] digest_0.6.39      R6_2.6.1           tidyselect_1.2.1   pillar_1.11.1     
## [33] magrittr_2.0.5     bslib_0.11.0       withr_3.0.2        tools_4.5.3       
## [37] gtable_0.3.6       cachem_1.1.0
```